Advanced Data Transformation Using Power Query in Power BI

Data transformation is a foundational step in Power BI that ensures the data is clean, structured, and ready for reporting. In many organizations, raw data—whether it comes from Excel files, text files, databases, or other sources—often arrives in formats not immediately suitable for analysis. Power Query provides a powerful set of features to restructure, clean, append, merge, pivot, unpivot, and enrich such data.

It provides a detailed walk-through of a real-world transformation scenario involving two Excel worksheets: one containing **revenue** data and the other containing **profit** data. Both datasets contain information for **five consecutive years** and **four regions** of a Pan-India organization. The goal is to convert these two separate worksheets with a wide layout (years as columns) into a single consolidated, analysis-ready dataset with a long layout (years as rows), including profit and revenue as separate fields.

This example demonstrates several advanced Power Query capabilities:

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- Connecting to multi-sheet Excel sources
- Promoting headers correctly
- Standardizing column names
- Appending multiple datasets
- Creating custom columns for category differentiation
- Unpivoting year columns
- Pivoting variable values into new fields
- Reordering columns for a custom template
- Applying and loading query steps

Each step is essential for restructuring data in a way that aligns with business requirements and supports efficient reporting.

1. Understanding the Raw Data Structure

Before beginning any transformation, it's important to understand the structure of the source data.

1.1 Source Workbook Structure

The Excel workbook contains two worksheets:

1. **Revenue** – Revenue values for each region across five years
2. **Profit** – Profit values for each region across the same five years

Both sheets share a similar layout:

- Region names appear in rows
- Year values (e.g., 2018, 2019, 2020, 2021, 2022) appear as column headers
- Data entries represent either revenue or profit

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However, the layout poses several challenges for easy reporting:

- Both sheets store **different measures (Profit/Revenue)** in separate sheets
- Years are arranged **horizontally as columns**, not vertically
- Headers are not automatically recognized upon import

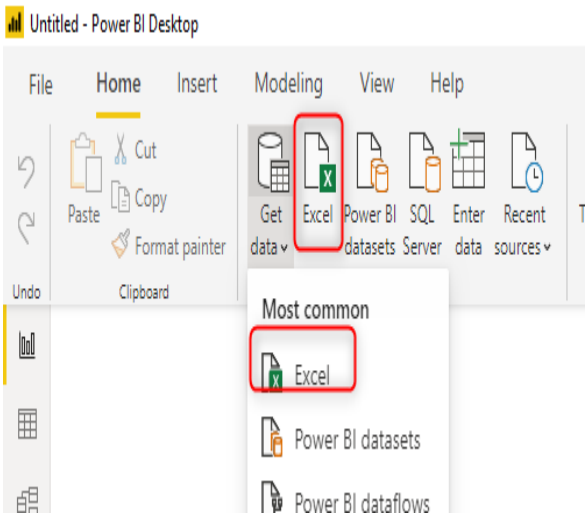
As a result, restructuring is required before analysis.

	A	B	C	D	E	F	G	H
1	**Revenue**							
2			2018	2019	2020	2021	2022	
3	North		100	120	130	150	160	
4	South		80	90	100	110	120	
5	East		60	70	80	90	100	
6	West		50	60	70	80	90	
1	**Profit**							
2			2018	2019	2020	2021	2022	
3	North		70	80	90	100	110	
4	South		40	50	60	70	80	
5	East		30	40	50	60	70	
6	West		20	30	40	50	60	

2. Loading Multi-Sheet Excel Data into Power BI

To begin the transformation:

1. Open Power BI Desktop and create a **new blank report**.
2. Click **Get Data → Excel Workbook**.
3. Choose the file named **structure_change_data.xlsx**.
4. Power BI opens a **Navigator window** showing:
    - **sheet1: Revenue**
    - **sheet2: Profit**
5. Select both checkboxes and click **Transform Data** instead of **Load**, to enter the Power Query Editor.



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This is the stage where all data transformation steps are applied before loading into the data model.

3. Fixing Incorrect Headers Using Promote Headers

When the sheets load into Power Query, Power BI often fails to detect the correct column headers, instead assigning generic labels like:

- Column1
- Column2
- Column3
  and so on.

To resolve this:

1. Select the **Revenue** table.
2. Go to **Home → Use First Row as Headers → Use Headers as First Row / Use First Row as Headers**.
3. Repeat the same for the **Profit** table.

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After promoting headers:

- Regions appear in the first column
- Year names appear correctly as column headers

If the header row contains additional descriptive rows, multiple promotions may be needed.

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4. Renaming Incorrect or Misidentified Columns

Even after promoting headers, Power BI may still misinterpret column titles.
For example, the first column may be labeled **Year** instead of **Region**.

To fix this:

1. Right-click the first column header.
2. Select **Rename**.
3. Enter **Region**.

Repeat this correction in both tables (Revenue and Profit).
Now both tables share a consistent structure, which is crucial for appending later.

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5. Creating a Category Indicator Using Custom Column

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Before appending data from the Revenue and Profit tables into one consolidated dataset, we must create a column that identifies whether each row represents **Revenue** or **Profit**.

Why?
Because after combining, the rows will no longer be distinguishable unless a variable field is added.

5.1 Adding a Custom Column in the Profit Table

1. Select the **Profit** table.
2. Go to **Add Column → Custom Column**.
3. In the dialog box:
    - **Column Name:** Variable
    - **Formula:** "Profit" (text in quotes)
4. Click **OK**.

A new column appears where every row is labeled **Profit**.

5.2 Adding a Custom Column in the Revenue Table

1. Select the **Revenue** table.
2. Repeat the custom column creation.
3. Set:
    - **Column Name:** Variable
    - **Formula:** "Revenue"

Consistency is essential:

- Column name **Variable** must match EXACTLY across both tables
- Spelling and capitalization must be identical

This ensures the column can later be used for pivoting.

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6. Appending the Two Tables into One Consolidated Table

Appending means stacking row-level data from multiple tables into a single dataset with identical columns.
It is different from merging (which joins tables horizontally).

Because the **Profit** and **Revenue** tables now share:

- The same column names
- The same number of columns
- A Variable column indicating category

They are ready to be appended.

6.1 Why Use Append Queries As New?

There are two options:

1. **Append Queries** – Appends data into the currently selected table
2. **Append Queries as New** – Creates a brand-new table with combined results

The recommended option is **Append Queries as New** because:

- The original datasets remain untouched
- Raw data integrity is preserved
- Debugging becomes easier
- Reusability of original queries is maintained

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6.2 Steps to Append

1. Go to **Home → Append Queries → Append Queries as New**.
2. In the dialog:
    - Choose **Two Tables** option
    - Table 1: Revenue
    - Table 2: Profit
3. Click **OK**.

Power Query generates a new table named **Append1** (or similar).
This table now contains:

- All regions
- All years
- Both Profit and Revenue values

- A Variable column to distinguish them

However, this table is still wide (years as columns), and needs restructuring.

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7. Unpivoting Year Columns (Converting Columns into Rows)

The appended table has five year columns, such as:

- 2018
- 2019
- 2020
- 2021
- 2022

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But the required structure needs:

- A single **Year** column
- One row per (Region, Year, Variable) combination

This requires **Unpivoting**, which is used when you want to transform multiple header columns into row values.

7.1 What is Unpivot?

Unpivot converts:

- Column headers → Row values
- Multiple year columns → One Year column

This is essential in long-format database design.

7.2 Steps to Unpivot Year Columns

1. Select the five year columns (Ctrl + Click).
2. Go to **Transform → Unpivot Columns → Unpivot Only Selected Columns**.

Power Query creates:

- A new column named **Attribute** (the year)
- A new column named **Value** (the numeric values)

Rename:

- **Attribute → Year**

At this point, the structure looks like:

| Region | Variable | Year | Value |

Where "Value" represents either profit or revenue depending on the Variable field.

This is called a **long format** or **tidy format** dataset.

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8. Pivoting the Variable Column (Creating Separate Profit and Revenue Columns)

After unpivoting, we have:

- Region
- Year
- Variable (Profit/Revenue)
- Value

But the final requirement is:

| Year | Region | Profit | Revenue |

This requires converting the Variable column (Profit/Revenue) into separate columns via **pivoting**.

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8.1 What is Pivot?

Pivot transforms:

- Row values → Column headers
- A categorical variable → Multiple new columns

In this case:

- Values in the **Variable** column become new columns
- Data in the **Value** column is placed accordingly

8.2 Steps to Pivot the Variable Column

1. Select the **Variable** column.
2. Go to **Transform → Pivot Column**.

3. In the dialog:
    - **Values Column:** select **Value**
4. Click **OK**.

Power Query creates:

- A **Profit** column
- A **Revenue** column

Both aligned with Region and Year.

9. Reordering Columns to Match the Required Template

According to the manager's required output structure, the columns must appear in this order:

1. **Year**
2. **Region**
3. **Profit**
4. **Revenue**

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To reorder columns:

- Drag and drop column headers
  or
- Use **Move → Left/Right/Beginning/End** options

This ensures the dataset is in the exact layout requested.

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10. Data Type Verification

Before loading the dataset:

- Region → Text (ABC)
- Year → Whole Number (1 2 3)
- Profit → Decimal or Currency
- Revenue → Decimal or Currency

Correct data types ensure proper aggregation and modeling.

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11. Applying and Loading the Query

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Once all transformations are complete:

1. Click **Close & Apply**.
2. Power Query applies each step sequentially.
3. Data loads into Power BI's model.
4. Relationships, if needed, are automatically detected.

The final consolidated dataset becomes available for charts, measures, dashboards, and further analysis.

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12. Summary of Key Concepts Used

This transformation exercise illustrates several essential data preparation skills:

12.1 Promote Headers

Fix unrecognized header rows.

12.2 Renaming Columns

Ensure consistency across tables.

12.3 Add Column → Custom Column

Create metadata fields for later differentiation.

12.4 Append Queries

Combine tables with identical structures.

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12.5 Unpivot Columns

Convert wide-format data into long-format data.

12.6 Pivot Columns

Convert categorical row values into new columns.

12.7 Column Reordering

Meet specific reporting or template format requirements.

12.8 Close & Apply

Load transformed data into Power BI Desktop.

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13. Final Notes on Scalability and Reusability

One major advantage of Power Query is its refreshability.

Once the transformation logic is built:

- Any new data added to the Excel sheets
- Any change in number of years
- Any update in values

can be automatically processed by clicking **Refresh**.

The steps dynamically adapt to changes:

- New years will appear as additional rows
- Additional regions will be automatically appended
- Profit/Revenue will remain separated

This makes the process highly scalable and efficient for ongoing reporting.

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14.Conclusion

This detailed transformation scenario demonstrates how Power Query empowers users to reshape raw, multi-sheet datasets into structured, analysis-ready tables. Using a blend of append, pivot, unpivot, custom column creation, header correction, and column reordering, complex data restructuring becomes quick, flexible, and repeatable.

By mastering these features, users can efficiently convert any irregular or wide-format data into meaningful long-format datasets, enabling smoother modeling and more accurate visualization in Power BI. This workflow is not only applicable to the profit–revenue example but is a universal pattern for preparing many real-world datasets.